A COMPARATIVE STUDY OF TRANSFER LEARNING FOR DRUM AUDIO STYLE CLASSIFICATION

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ABSTRACT

Research in drum classification often trends in two key directions; (1) audio based single instrument classification, or (2) automatic drum transcription which makes exploration of drum audio style compelling, especially in comparison to a CNN trained on the GMD (Groove Midi Dataset) [1] with a pretrained transformer based model PaSST (Patchout Audio Spectrogram Transformer) [2] with frozen general audio embeddings from AudioSet. This comparison reveals the ways in which general audio knowledge can affect drum audio style classification. Experiments with model depth, augmentation and padding show that PaSST with these frozen embeddings reduces performance in terms of accuracy but reveals robust feature representation distinct from the CNN.

1. INTRODUCTION

Prior research on drums has focused either on single instrument identification [3] or symbolic pattern analysis [1, 2], leaving a gap in audio-based style recognition. A drum style can be understood as a semantic label designated for characteristic patterns and timbres over time making the Groove MIDI Dataset (GMD) [3], one of the largest drum audio datasets with hierarchical style labels, ideal for the project. Transfer learning from general audio corpora can yield benefits in downstream tasks [4–8] and CNNs have proven effective for classifying individual drum instruments in limited data settings [9]. Additionally, pretrained transformer-based models have achieved SOTA results in general audio and music tagging tasks [8, 10]. In particular the PaSST model; a lightweight Audio Spectrogram Transformer [11–13] with patchout regularization [14] has demonstrated consistent gains over CNNs in instrument recognition and music genre classification. This led to the research question; What is the effect of pretrained general audio embeddings in PaSST when compared to a baseline CNN for drum audio style classification?

2. METHOD

To investigate this a CNN trained solely on GMD is compared against PaSST using frozen AudioSet [15] embeddings. By examining both accuracy and model interpretability, the effect of the pretrained embeddings in contrast with a CNN can be more clearly understood. The GMD contains 18,264 2-bar recordings with primary and secondary style annotations yielding 74 unique style classes upon concatenation. GMD audio is used in its default 16 kHz mono for the CNN and resampled to 32Khz mono for PaSST then padded to fixed length of 10 seconds to match AudioSet embeddings and ensure fair comparison between models. Mel spectrograms are computed with training, validation, and test sets of approximately 80/10/10 percent with proportional class balancing ensuring each split has the same relative proportion of each class. Performance is evaluated with F1-score as the primary metric.

The CNN is a VGG (Visual Geometry Group) inspired model adopted from Hiner with experiments conducted to optimize convolutional layers [16]. PaSST uses a lightweight AST-based architecture [14] without fine-tuning to isolate the effect of the pretrained embeddings on the target task. On top of this an MLP (Multilayer Perceptron) is refined through experiments with depth and bottlenecks. Both architectures are trained with the Adam optimizer with a learning rate of 1e-4 and batch size of 16 for 50 epochs with early stopping patience of 10. Once model depth is optimized, experiments with augmentations [4] GaussianNoise, RoomSimulator and TimeStretch commence, followed by comparing padding modes (zeros, reflection, circular) to assess their impact on feature robustness as these techniques have been employed to enhance generalization in audio classification tasks [17–19]. For interpretability, t-distributed stochastic neighbor embedding (t-SNE) is utilized which reveals differences in clustering between architectures while the confusion matrices [5] identify differences in performance.

[1] https://www.tensorflow.org/datasets/catalog/groove#groove2bar-16000hz
[2] https://github.com/kkoutini/passt_hear21
[3] https://github.com/khiner/DrumClassification

[4] https://github.com/iver56/audiomentations
[5] https://drive.google.com/drive/folders/1taQUOIU8z7aKlKEZvoMsV9u3fP7IBS_C?usp=drive_link

3. RESULTS

The CNN maintained a higher F1-score across a total of 34 experiments in 11 rounds within four categories; Configuration Exploration (rounds 1-2), Dataset Exploration (3-5), Model Depth (6-9), Augmentations and Padding (10-11) [5]. As shown in Table 1 The best CNN (7-layer) attained F1 of 0.8944, whereas the best PaSST configuration (4-layer MLP) reached F1 0.8659. When using a 10 percent subset of GMD (GMD-mini), PaSST outperformed the CNN (F1 0.3911 vs 0.3267), suggesting the pretrained AudioSet embeddings confer an advantage in data-sparse conditions. RoomSimulator and GaussianNoise augmentations slightly boosted CNN performance but degraded PaSST's accuracy, while TimeStretch hindered both models' performance. The CNN with augmentations ultimately achieved the highest score (F1 0.9080). These results demonstrate that frozen general audio features alone did not yield an accuracy improvement over the CNN although they were beneficial when data was limited. Both architecture's confusion matrices indicate struggles with similar classes [5]. For instance, both the CNN and PaSST frequently confuse 'jazz-fast' grooves with 'dance-breakbeat' and 'jazz-mediumfast'. Likewise, certain latin styles were confused by both models with funk versus hip-hop showing confusion. The CNN was notably prone to over-predicting primary style 'rock' which is likely attributed to class imbalance as rock styles compose nearly two-thirds of the GMD. With augmentations the CNN became better at distinguishing similar funk and jazz classes, while augmentations with PaSST introduced confusion among latin and funk classes along with patchout augmentation on spectrograms providing no clear advantage.

The learned feature representations of both models differ markedly with Figure 1 illustrating PaSST's embeddings with distinct clusters corresponding to primary style indicating the pretraining helps build a more hierarchical representation of drum audio style. In contrast, Figure 2 shows the CNN's latent space which is characterized by smaller clusters of individual style classes. This suggests that PaSST's AudioSet embeddings include information relevant to an underlying representation of drum audio style. PaSST demonstrates a robust embedding with minimal mean centroid shift until reflection padding at 8.32 among primary styles with accuracy notably reduced through waveform augmentations revealing similarly confused classes providing further insights about its representation of primary style. The CNN displays over 10x mean centroid shift at 94.66 by primary style from reflection padding, with TimeStretch augmentation revealing its sensitivity to class imbalance along with a peculiar 'misfits' cluster in the latent space that may be composed of drum fills (291 of the original files were labeled 'fill' rather than all others as 'beat') or more varied transients.

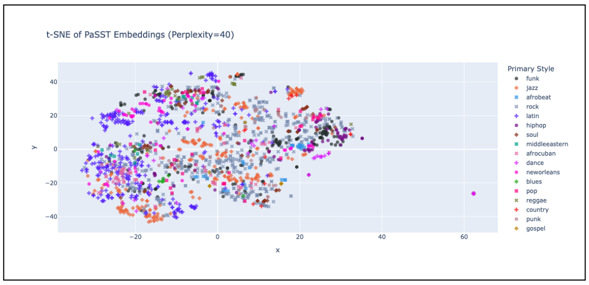

Figure 1. PaSST t-SNE by primary style.

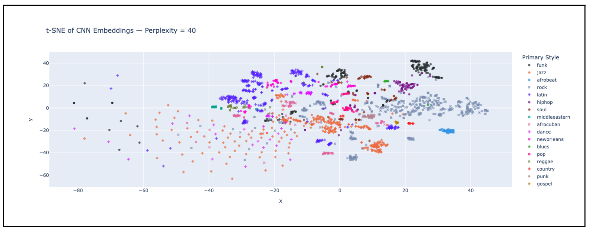

Figure 2. CNN t-SNE by primary style.

4. CONCLUSION

In terms of accuracy the CNN slightly outperformed PaSST however with a lesser degree of broader primary style based feature understanding compared to PaSST hinting at more adherence to local rather than global features. From an interpretability standpoint, PaSST appears to learn a more hierarchically structured embedding space evidenced by lower mean centroid shift and tighter clustering by primary style. This implies the 'world model' from AudioSet contained within PaSST was beneficial in terms of understanding drum audio style even in the presence of lower overall accuracy. Further analyses highlight that both models confuse similar styles, while the CNN was more impacted by class imbalance in the training data despite proportional balancing across splits. PaSST more readily maintained stable clusters under GaussianNoise, RoomSimulator and TimeStretch augmentations that at times lowered accuracy. Future research can explore finetuning PaSST or other transformer based models on GMD, ensemble models for auto tagging of polyphonic recordings and applications to real world drum audio recordings in semi-supervised or unsupervised contexts.

ID	Model	F1	Description
1.1	CNN	0.9204	Defaults with rock styles
2.2	PaSST	0.8699	PaSST config
3.2	PaSST	0.3911	GMD-mini
4.1	CNN	0.7531	GMD-full
5.4	CNN	0.7827	GMD-full repeat
6.1	PaSST	0.7247	t-SNE & Time patchout
7.1	PaSST	0.8582	PaSST MLP 3 Layers
8.3	PaSST	0.8659	PaSST MLP 4 Layers
9.2	CNN	0.8944	CNN 7 Conv Layers
10.1	CNN	0.9080	CNN 7 Conv Layers + Noise, RoomSim
11.2	PaSST	0.8752	PaSST MLP 4L, Reflection padding

Table 1. F1 scores for the best model per round.

5. REFERENCES

[1] R. Vogl, "Deep Learning Methods for Drum Transcription and Drum Pattern Generation," Ph.D. thesis, Inst. of Computational Perception, Johannes Kepler Univ. Linz, Austria, Nov. 2018.

[2] L. Géré, N. Audebert, and P. Rigaux, "Improved Symbolic Drum Style Classification with Grammar-Based Hierarchical Representations," in *Proc. 25th Int. Society for Music Information Retrieval Conf. (ISMIR)*, San Francisco, CA, 2024.

[3] J. Gillick, A. Roberts, J. Engel, D. Eck, and D. Bamman, "Learning to Groove with Inverse Sequence Transformations," in *Proc. 36th Int. Conf. on Machine Learning (ICML)*, Long Beach, CA, 2019.

[4] Q. Kong, Y. Cao, T. Iqbal, Y. Wang, W. Wang, and M. Plumbley, "PANNs: Large-Scale Pretrained Audio Neural Networks for Audio Pattern Recognition," *arXiv preprint arXiv:1912.10211v5*, 2020.

[5] K. Koutini, H. Eghbal-zadeh, and G. Widmer, "Receptive Field Regularization Techniques for Audio Classification and Tagging with Deep Convolutional Neural Networks," *arXiv preprint arXiv:2105.12395v1*, 2021.

[6] S. Chen, Y. Wu, C. Wang, S. Liu, D. Tompkins, Z. Chen, and F. Wei, "BEATS: Audio Pre-Training with Acoustic Tokenizers," *arXiv preprint arXiv:2212.09058v1*, 2022.

[7] A. Quelennec, M. Olvera, G. Peeters, and S. Essid, "On the Choice of the Optimal Temporal Support for Audio Classification with Pre-Trained Embeddings," *arXiv preprint arXiv:2312.14005v1*, 2023.

[8] Y. Ding and A. Lerch, "Audio Embeddings as Teachers for Music Classification," in *Proc. of the 24th Int. Society for Music Information Retrieval Conf. (ISMIR)*, Milan, Italy, 2023.

[9] M. Yun and J. Bi, "Deep Learning for Musical Instrument Recognition," Tech. Report, Dept. of Electrical and Computer Eng., Univ. of Rochester, 2018.

[10] K. Choi, G. Fazekas, M. Sandler, and K. Cho, "Transfer Learning for Music Classification and Regression Tasks," in *Proc. 18th Int. Society for Music Information Retrieval Conf. (ISMIR)*, Suzhou, China, 2017.

[11] Y. Gong, Y.-A. Chung, and J. Glass, "AST: Audio Spectrogram Transformer," *arXiv preprint arXiv:2104.01778v3*, 2021.

[12] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training Data-Efficient Image Transformers Distillation through Attention," *arXiv preprint arXiv:2012.12877v2*, 2021.

[13] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An Image Is Worth 16×16 Words: Transformers for Image Recognition at Scale," *arXiv preprint arXiv:2010.11929v2*, 2021.

[14] K. Koutini, J. Schlüter, H. Eghbal-zadeh, and G. Widmer, "Efficient Training of Audio Transformers with Patchout," *arXiv preprint arXiv:2110.05069v3*, 2022.

[15] J. F. Gemmeke, D. P. W. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, and M. Ritter, "AudioSet: An ontology and human-labeled dataset for audio events," in *Proc. IEEE ICASSP*, 2017, pp. 776–780.

[16] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," *arXiv preprint arXiv:1409.1556v6*, 2014.

[17] A. Gazneli, G. Zimmerman, T. Ridnik, G. Sharir, and A. Noy, "End-to-End Audio Strikes Back: Boosting Augmentations Towards an Efficient Audio Classification Network," *arXiv preprint arXiv:2204.11479v5*, 2022.

[18] H. Park, Y. Chung, and J.-H. Kim, "Deep Neural Networks–based Classification Methodologies of Speech, Audio and Music, and its Integration for Audio Metadata Tagging," *J. Web Eng.*, vol. 22, no. 1, pp. 1–26, Apr. 2023.

[19] T. Morocutti, F. Schmid, K. Koutini, and G. Widmer, "Device-Robust Acoustic Scene Classification via Impulse Response Augmentation," *arXiv preprint arXiv:2305.07499v2*, 2023.